# AMATH 403 HW 5

Jordan Tucker

May 11, 2026

---

**1: 4.3.18. Use separation of variables to solve the Helmholtz boundary value problem $\Delta u = u$, $u(x, 0) = 0$, $u(x, 1) = f(x)$, $u(0, y) = 0$, $u(1, y) = 0$, on the unit square $0 < x$, $y < 1$.**

---

**Using separation of variables:**

$$u(x, y) = v(x)w(y)$$

**Helmholtz equation:**

$$\Delta u = u$$

$$u_{xx} + u_{yy} = u$$

**Using substitution:**

$$u_{xx} = v''(x)w(y)$$

$$u_{yy} = v(x)w''(y)$$

$$v''(x)w(y) + v(x)w''(y) = v(x)w(y)$$

**Divide by** $v(x)w(y)$